

Harold Andrey Martínez Cortés

AI & Automation Engineer

I design and implement artificial intelligence solutions that automate processes, accelerate data analysis and strengthen the operations of companies and growing businesses: from reports generated by AI agents to custom systems for web, desktop, mobile and hardware.

EXPERIENCE

AI Data Analytics Solutions Consultant
Mar 2026 — Present

Independent consultant · Client working with a US company

- Cut the turnaround time of reports and dashboards from **2–3 weeks to 1–2 days** with generative AI workflows.
- Complete reports generated from **a single instruction** (one-shot), in **4 formats**: Excel, PowerPoint, Power BI and Datorama.
- Developed AI agents that produce the **weekly reports** and specific projects on a schedule.
- Integrated the Claude API, Claude Code and Cowork with Power BI, and automated processes with Power Automate and Azure.

Custom Software Developer
2026

Independent projects

- GymApp, **in production with 300+ registered clients**: memberships, point of sale, inventory, cash register and biometric access control with **3 ways in** (fingerprint, card or PIN) and automatic door opening via Arduino Nano.
- Gym web platform with a commercial site and a private area with **4 staff roles** (developer, admin, front desk and trainers) and personal access for every client.
- Mis Finanzas: Android personal finance app with **7 modules**, visual analytics and automatic recommendations; all data stays on the phone.
- AI-assisted development (Claude Code) across the full cycle —design, implementation and testing—, with **27 automated test suites** in GymApp.

Electronics Engineering, Automation & Networking
2020 — 2026

Academic and independent projects

- Designed and built mini-sumo robots, audio amplifiers and printed circuit boards (PCB).
- Home automation systems with web control interfaces; IoT prototypes with ESP32, camera, microphone and temperature, humidity and voltage sensors.
- Network installation and configuration: routers and structured cabling.

SELECTED PROJECTS

AI Reports AI & data — View project ↗

GymApp Desktop application — View project ↗

Planillero Vóley Mobile app — View project ↗

BiciFest Colombia 2026 Web platform — View project ↗

Moton Web platform — View project ↗

See all 9 projects in the portfolio ↗

CONTACT

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EDUCATION

B.S. in Electronics Engineering
Universidad de Cundinamarca
2020 — 2026

Specialization in Network Infrastructure and Security
Universidad de Cundinamarca
2025 — 2026

CERTIFICATIONS

AI Engineer
Udemy · Platzi · Netzum
2026

Prompt Engineering
Udemy · Platzi · Netzum
2026

Claude & agents courses
Anthropic
2026

SKILLS

AI & language models
Claude (API, Code, Cowork) · ChatGPT · Gemini · Copilot · Ollama · OpenCode · OpenClaw · Prompt engineering · AI agents

Data & reporting
Power BI · Power Automate · Power Apps · Copilot Studio · Excel · PowerPoint · Datorama · SQL · Python · Azure

Software development
JavaScript · TypeScript · HTML/CSS · React · Next.js · Node.js · Electron · Android Studio · Capacitor · SQLite

Tooling & deployment
Git · GitHub · Vercel · Supabase · Vite · Tailwind CSS

Electronics & IoT
Arduino · ESP32 · Sensors · Cameras & audio · PCB design

Networking & security
Network infrastructure · Network security · Routers · Structured cabling